Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/28/2013
PI Name:	Paul, Anna-Lisa Ph.D.		
Project Title:	The biological impact of spaceflight and extraterrestrial environments on molecular signaling and gene expression in plants		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32611-0690	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	06/09/2009	**End Date:**	06/08/2013
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 6/8/2013 per PI/NSSC info (Ed., 4/20/2012)		
Key Personnel Changes/Previous PI:	Agata Zupanska - Postdoctoral Research Associate ; Fiona Denison - Postdoctoral Research Associate ; Matthew Bamsey - Postdoctoral Research Associate ; Claire Amalfitano - Biological Scientist ; Lawrence Rasmussen - Biological Scientist		
COI Name (Institution):	Ferl, Robert (University of Florida)		
Grant/Contract No.:	NNX09AL96G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The overall goal of our program and this proposal is to understand the fundamental molecular responses of plants to spaceflight environments, with particular emphasis on the changes in gene expression and signal transduction that accompany spaceflight experience. We have used genetically engineered plants and molecular tools to monitor plant responses to spaceflight and space-related analog environments and have developed specific hypotheses regarding plant signaling responses that are affected by spaceflight. The focus of this proposal is to build on these extensive insights to develop the next generation of biological assays and hardware concepts to address these hypotheses in ground and analog studies. In the process, these new biological and analytical tools will be integrated into functional concepts than can be readied for future deployment in a range of spaceflight-related opportunities. The biology will include new biosensors to test hypotheses regarding the mechanisms by which molecular signals are conveyed in response to spaceflight and analog environments. Some biosensors will be linked to Green Fluorescent Protein variants to facilitate multiplexed telemetric data collection. The analytical techniques will make use of current spaceflight imaging technologies and explore advanced imaging concepts. This combination of biosensors and imaging will enable robust robotic experiments with flexible future deployment options, enabling experiments that address science needs to survey molecular responses in a variety space venues and situations. The expected significance of this project is that it will advance the fundamental understanding of spaceflight molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected significance of this project is that it will advance the fundamental understanding of molecular signaling in spaceflight-relevant analogs and then provide the means to transition these studies for tests in bona fide spaceflight environments as opportunities develop. Thus the objectives of this proposal are completely aligned with the request of the NRA to investigate the nature of cellular receptors, signal transduction pathways and changes in gene expression in response to spaceflight-relevant environments. The concomitant ground and flight support represented by this type of grant enables us to combine the processes of flight biology preparation, flight data analysis, and hypothesis development and testing for unprecedented insight into biological responses to these novel environments. The execution of this grant was (and continues to be) the ideal example of what NASA intended as the role of Fundamental Space Biology Ground-Based Research: NRA-01-OBPR-06 "…to expand our understanding of fundamental biological processes and the mechanisms by which these processes sense, respond, adapt to, and evolve in the space environment…and in particular… the role of the genome and cellular structures in sensing and responding to gravitational force."
	Overview and Objectives – a Synopsis During spaceflight, plants experience growth and developmental conditions that are distinctly unlike the terrestrial environment that has directed the evolution of gene expression patterns that ensure survival. Plants must therefore interpret novel stimuli, integrate new solutions and express potentially unusual suites of genes in an attempt to cope with spaceflight environments. Some of these responses might be directly interpretable as appropriate responses that clearly inform science of the stresses induced by spaceflight. Other responses might be confused or inappropriate due to the novel signals or biological processes present in the milieu of the spaceflight environment. Defining both kinds of responses is critical to understanding biological response and adaptation to all kinds of extraterrestrial environments, from vehicles in orbit to lunar habitats. Our objective is to answer fundamental questions regarding the metabolic and molecular response of plants to novel, spaceflight relevant, environments. In the current project we build on extensive insights gained from spaceflight and analog experiments to develop new biological tools and to guide hardware development to study the response to these environments with higher fidelity and wider scope. The biology will include new biosensors created to dissect the mechanisms by which molecular signals are conveyed and integrated into responses to spaceflight environments. Some biosensors will be linked to Green Fluorescent Protein (GFP) spectral variants to facilitate telemetric data collection, which will in turn enable robust robotic experiments with wide deployment options. The hardware to support this telemetric science, will be advanced in concept to include multiple spectral capacities in order to monitor multiple biosensor reporter genes simultaneously. The focus is on advancing telemetric imaging that will enable the potential future deployment of such biosensor-units in a wide range of potential spaceflight applications - ISS, crew vehicles, satellites and lunar or Mars packages - as the opportunities present. These experiments address the need to understand the biological response of eukaryotes to spaceflight, and other novel environments at the molecular and cellular level, with special focus on signaling pathways and gene expression. Summary Focus and Opportunities The final year of the grant continued to help support people and efforts for additional characterizations of returned Flight materials from the PIs' two Flight Experiments: APEX-TAGES and BRIC-16, as well as contributing to the support of ancillary analyses associated with the BRIC-17 returned material. Space Biology support enabled an additional level of analyses of these samples, and facilitated supplemental ground studies that continue to contribute to our understanding of the response of biology to spaceflight and novel environments. The data analyses led to the design and support of several analog studies initiated last year: 1) clinorotation and gravity vector disruption, including root skewing analyses in inclined planes, 2) hypergravity tests with a 2-3g centrifuge, and 3) a parabolic flight campaign opportunity and suborbital flight preparations in high performance aircraft (in support of two FO awards). The concomitant ground and flight support represented by these grants enables us to combine the processes of flight biology preparation, flight data analysis, and hypothesis development and testing for unprecedented insight into biological responses to these novel environments. The execution of this grant was (and continues to be) the ideal example of what NASA intended as the role of Space Biology Ground-Based Research: NRA-01-OBPR-06 "…to expand our understanding of fundamental biological processes and the mechanisms by which these processes sense, respond, adapt to, and evolve in the space environment…and in particular… the role of the genome and cellular structures in sensing and responding to gravitational force."

Task Progress:	The final year of the grant had several foci: Undifferentiated cells - cell sensing and signal transduction: The results of the BRIC-16 experiment (see FSB 2012 report and related BRIC-16 report) revealed that undifferentiated cells can detect the spaceflight environment and altered gravity in the absence of differentiated tissue or organized developmental structures (Paul et al. 2012: BMC Plant Biology. 12:232 http://www.biomedcentral.com/ , Zupanska et al. 2013: American Journal of Botany. 100:235-248). One of the most striking results was the engagement of "heat shock" and protein chaperone genes The results from the tissue culture experiments leads us to further dissect the sensory and signal mechanism(s) that cell cultures use to detect and respond to spaceflight. To further address these questions, we started a series of ground experiments to test signal transduction pathways in unit gravity, and to also stage out biological tools for future flight opportunities. We initiated several new tissue culture lines, some of them containing GFP reporters, and others that are deficient in key chaperone genes. Experiments with cultures carrying known mutations that are well characterized in the gravity sensation and signaling of intact plants will provide insight into the mechanisms behind those processes. If cell cultures are using these typical sensing and signaling elements, the mutations should greatly attenuate the cell culture responses Reporter gene line development: The results of the APEX-01 / TAGES and the BRIC 16 experiment revealed that a large number of genes were differentially expressed in response to the spaceflight environment (Paul et al. 2012: BMC Plant Biology. 12:232 http://www.biomedcentral.com/ ; Paul et al 2013: BMC Plant Biology. 13:112 http://www.biomedcentral.com/). These results initiated the development of a variety of new GFP-reporter gene lines to support spaceflight applications. Biosensors in calcium-signaling mutant backgrounds: A number of reporters were developed in calcium signaling mutant backgrounds; two lines show differential expression of the reporter gene in plants deficient in aspects of calcium signaling. Quality testing of flight hardware: This grant helped to support the bio-compatibility testing of two types of spaceflight hardware under consideration, the Aluminum PDFUs for future BRIC experiments, and a new ceramic foam sponge for the ABRS plant biology growth chamber. Parabolic flight campaigns (Zero-G): This Space Biology grant contributed to the support of several parabolic flight campaigns that were awarded through the Flight Opportunities Office. Since FO does not provide operations support, it is essential that the PIs have some other source of NASA funding to support the personnel and analyses. Our NASA NNX09AL96G grant was the foundation of this support, and crucial for our success. Performance aircraft for Sub-orbital application: The other un-supported Flight Opportunity Award received by the PIs is for a series of suborbital flights. This Space Biology grant also contributed to the support of preparations for the suborbital FO by conducting hardware and biology tests in high performance aircraft (T6 and F104). Concluding remarks: We have accomplished an amazing amount of diverse, spaceflight-related science with this grant. Having this support has allowed us to develop a large range of biological and engineering tools to support four flight experiments, three parabolic flight campaigns, and enhance the technical readiness of our upcoming suborbital flight opportunity – all while conducting novel, fundamental ground-based research that expands the communities understanding of the molecular mechanisms associated with the spaceflight acclimation of plants. This research funding contributed to 16 publications over its lifetime, and to the support of three PhD students (Anne Visscher, Eric Schultz, Natasha Sng), three undergraduates (Lauren Frizzle, Justin Goodwyn, Zackary Guignardi), four postdoctoral associates (Agata Zupanska, Fiona Denison, Matthew Bamsey, Thomas Graham), and two biological scientists (Claire Amalfitano, Lawrence Rasmussen).
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